Problems and physics

1. Problem Statement

The official problem addressed by DRISHTI is the development of an:

Requirement	What our system does	Status
Gather SSS data and location in formation	Analyze raw Side-Scan Sonar(SSS) and location file	Done
Remove noise and handle image problems	Reduce noise,improves image contrast and handles missing sonar data	Done
Detect underwater man made objects	Use YOLOv8s to detect four types of objects	Done
Give reliable confidence scores	Gives each detection a confidence percentage and check’s the objects shadow to reduce false reduction	Done
Find the location and size of objects	Convert sonar image positions into latitude, longitude and object dimension	Done
Generate detection reports	Creates reports in JSON and CSV format	Done
Provide a dashboard	Provide react,leaflet map dashboard for uploading data viewing detections and downloading reports	In progress
Run of edge devices	Uses lightweight ONNX and cpu setup	done

A sonar survey can cover a large area in a relatively short period. The project documentation estimates that a survey vessel can map approximately 130 hectares in a day. However, the resulting imagery still has to be examined. A major problem is that most of the imagery represents ordinary seabed.This means that an analyst spends a considerable amount of time looking at areas where there is nothing important.Another issue is human fatigue. Long periods of manually reviewing similar-looking sonar imagery can make it difficult to notice small or weak objects.The problem becomes more difficult because natural formations can sometimes resemble artificial objects.A ghost net can be very difficult to distinguish because of its weak acoustic response.Therefore, simply training an object detector is not enough. The system also needs to consider the physical characteristics of sonar imagery.

2. Understanding Side-Scan Sonar

Side-scan sonar is different from an ordinary camera.A camera records light reflected from objects. A side-scan sonar system sends acoustic pulses into the water and records the echoes that return from

the seabed and objects.The project documentation describes the sonar image as a map of how sound returns from the underwater environment.
Speckle is an inherent characteristic of coherent imaging systems.The project models the observed image approximately as:$I_{obs} = I_{true} \times n$
where:I_{obs} is observed image,I_{true} is underlying signal and n is multiplicative noise component.Because speckle affects image intensity, simple averaging can remove useful object boundaries along with noise.

Contrast enhancement can help make weak structures more visible.The project documentation identifies CLAHE as part of the preprocessing approach.However, enhancement must be controlled because excessive enhancement can also increase unwanted speckle.
Objects farther from the sonar generally appear weaker because acoustic energy decreases with distance.As a result, the same object may look different depending on its range.Without correction, an AI model could learn the distance-related brightness pattern instead of learning the actual object.

Shadow Geometry

The project uses the relationship:$L = h \times G / (H - h)$

where:L = expected shadow length,h = object height,H = sonar altitude above the seabed,G = ground range,R = slant range

The ground range can be calculated as:$G = \sqrt{R^2 - H^2}$

A simpler textbook form of the same relationship is:

$L = h / \tan(\theta)$

where θ represents the relevant grazing angle.

The project uses the range/altitude form because those quantities can be obtained from sonar information.

3. Types of Sonar Considered

Several sonar technologies can be used for underwater surveying.

Sonar Type	Main Purpose
Side-Scan Sonar	Surface texture and object detection
Forward-Looking Sonar	Forward visibility and obstacle detection
Synthetic Aperture Sonar	High-resolution acoustic imaging
Multibeam Echo Sounder	Depth and 3D bathymetry

1 Side-Scan Sonar:-Side-scan sonar scans across the vehicle's direction of travel and produces detailed information about the seabed surface.It is particularly useful for identifying:Shipwrecks,pipes,cylinders,debris,acoustic shadows.

2 Forward-Looking Sonar:-Forward-looking sonar observes the area in front of the vehicle.It is useful for navigation and obstacle avoidance but is not primarily designed for systematic seabed survey coverage.
3 Synthetic Aperture Sonar:-Synthetic aperture sonar combines information from multiple positions to achieve much higher resolution.

4 Multibeam Echo Sounder:-Multibeam systems measure depth across a wide area and are mainly used to create bathymetric maps.The project selects side-scan sonar because it provides both surface texture and acoustic-shadow information while being suitable for systematic surveying. 4. Target Objects and Their Acoustic Characteristics

4.Target classes

Different underwater objects produce different acoustic signatures.

This is important because the AI system cannot treat every target in exactly the same way.

1 Shipwrecks:-Shipwrecks generally produce strong acoustic returns because of their size and material.Their acoustic shadows are usually large and irregular.
Typical characteristics:-Strong backscatter,large shadow,irregular structure,significant elevation,Strong contrast with the surrounding seabedThe shadow may outline parts of the structure and provide information about the object's height and shape..

2 Pipes and Pipelines:-Pipes and cylinders are generally rigid objects.Their acoustic response can be strong and concentrated.Because of their shape and elevation, they can produce relatively straight and predictable shadows.
Typical characteristics:Strong backscatter,Linear appearance,Sharp shadow,Predictable geometry,Long continuous structure
This makes geometric and shadow-based checks particularly useful for this category.
3 Cylindrical Objects:-Cylindrical objects have acoustic properties similar to pipes but may have different sizes and aspect ratios.
Mine cylinders generally produce a moderate-to-strong compact sonar return.
They have relatively regular geometry and may produce a short, sharp acoustic shadow. Their distinctive aspect ratio can help separate them from other objects, although they may still be confused with pipe segments or rocks.
The project treats cylinders separately because the available mine-contact data has a distinct scale and aspect ratio and because the operational response to a possible ordnance-related detection is different.

4 Ghost Nets:-Ghost nets are considerably more difficult to detect.They are usually made from thin and flexible material such as nylon.Compared with metal or concrete objects, they may produce much weaker acoustic returns.Their shadows can also be fragmented or almost absent.
Typical characteristics:Weak backscatter,Irregular shape,Thin structures,Fragmented shadow,Difficult separation from seabed patterns
Because both the backscatter and shadow signals can be weak, ghost nets require a different treatment from larger rigid objects.The project therefore does not rely entirely on the shadow check for ghost nets.

5. How an Acoustic Map Is Made

A side-scan sonar does not take a normal photograph of the sea floor. Instead, it sends sound waves into the water and records the sound that comes back.The basic process is:
Sonar sends sound → sound reaches sea floor → sound is reflected → sonar receives the echo → computer converts the echo into an image
When the sound reaches the sea floor, different surfaces reflect different amounts of sound.
For example:A hard object can produce a strong return,a soft seabed can produce a weaker return,an object can block the sound behind it and create a dark acoustic shadow.
The sonar records these differences and creates an image called an acoustic map or side-scan sonar image.
A sonar transducer sends an acoustic pulse toward the seabed. The sound interacts with the seabed and objects present on it. Some of the acoustic energy is reflected back toward the sonar receiver.The receiver measures the returning signal.The time taken by the signal to travel to the target and return provides information about its distance from the sensor.For a simplified range calculation:Distance = (Sound speed × Travel time) / 2
The division by two is required because the measured time represents the complete journey of the sound: from the sonar to the target and back.Each sonar ping provides a cross-track measurement. As the sonar moves forward, consecutive pings are placed together to form the larger sonar image.

One of the most useful characteristics of side-scan sonar is the acoustic shadow.When an object rises above the seabed, it can block part of the acoustic beam.The area behind the object receives less or no direct acoustic energy and therefore appears darker.This dark region is called the acoustic shadow.The shadow contains information about the height and geometry of the object.For example, a large shipwreck may produce a large and irregular shadow, while a small cylindrical object may produce a narrower and more regular shadow.This makes the shadow useful for distinguishing artificial objects from natural formations.

6. Edge and Offline Deployment

A practical underwater survey system cannot always depend on a continuous internet connection.Marine drones and autonomous underwater vehicles may operate in environments where communication is limited or unavailable.Therefore, the system is designed to support offline processing.The project includes a Torch-free ONNX inference path using:ONNX Runtime,NumPy,OpenCV
The documented benchmark is approximately **90 ms per tile on CPU** for this inference path. This makes it possible to move inference closer to the survey platform rather than requiring every image to be transmitted to a remote server.

Training Classes and Dataset Strategy

The project uses a slightly different internal taxonomy from the original problem statement.

Problem Category	DRISHTI Class	Status
Shipwrecks	`shipwreck`	Product class
Pipes	`pipeline / cylinder`	Product class
Cylinders	`mine cylinder`	Product class
Entangled debris nets	`ghost_net`	Synthetic-trained
—	`crab_pot`	Hard negative
—	Empty seafloor	Background